SPE-227186-MS

Development of Non-Iron Crosslinked Self-Diverting Gelled Acids for Carbonate Acidizing

N. Nurlybayev, M. Farouk, A. Yudin, and Z. Al Jalal, TAQA, Saudi Arabia; H. M. Li, K. K. Cheng, Y. Wang, and Z. J. Xiao, OPT, China

Copyright 2025, Society of Petroleum Engineers DOI 10.2118/227186-MS

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

It is often necessary to apply diversion techniques in the acid treatment procedures to ensure homogeneous coverage of treatment zones and maximize the acidizing effectiveness. Due to the heterogeneous properties of carbonate formations, it is even more critical to design diversion properly in carbonate acidizing than sandstone acidizing. This paper will describe the process of developing a special non-iron crosslinked self-diverting gelled acids for efficient carbonate acidizing.

During acidizing operations, acids tend to flow into less flow resistance areas because these areas have higher permeability or less damage, leaving the low permeability areas non-treated. In order to control the placement of treatment fluids, a number of diversion techniques have been developed for carbonate acidizing treatment, both mechanical and chemical. Although mechanical techniques such as inflatable or straddle packers and ball sealers are very effective, they are more expensive and operationally time consuming than chemical diversion techniques, and furthermore not applicable in open hole and gravel packed well completions.

The study identifies the non-iron crosslinked self-diverting gelled acids as the most cost-effective chemical diverter which are self-diverting and less damaging. There are mainly two types of polymeric self-diverting gelled acids, iron-crosslinked and non-iron crosslinked. The ferric ions in iron-crosslinked self-diverting gelled acids are chemically active and tend to form precipitations and create secondary formation damage especially in treatment of sour gas or oil reservoirs. Therefore, it is necessary to develop non-iron crosslinked self-diverting gelled acids in acidizing treatment of carbonate reservoirs. The results indicate that novel non-iron crosslinked self-diverting gelled acids, both fresh and spent acids, are stable at atmospheric and bottom hole conditions. Minimal corrosion (PI≤1, corrosion rate≤0.05 lbs/ft^2) was obtained by introducing the unique corrosion inhibitor package. Up to 3000 ppm ferric ion was controlled properly in fluid design. The viscosity buildup of spent acids at pH 2-3 increases the fluid flowing pressure and provides effective diversion. The de-crosslink mechanism at higher pH (4-5) breaks the acidic gels, leaving less damage to the formation.

Downloaded from https://onepetro.org/SPEMEOS/25MEOS/conference/25MEOS by JSC Zarubezhneft, Alexey Solovyev on 07 November 2025

This paper describes the development of non-iron crosslinked self-diverting gelled acids. A novel non-iron crosslinking chemical mechanism is introduced for the first time. Fluid properties of diversion, stability, iron control, corrosion and rheology are evaluated in acidizing laboratory.

Introduction

The carbonate rocks constitute approximately 20% of sedimentary rocks worldwide. Furthermore, carbonate reservoirs contain more than 70% of the world's oil and gas resources, including approximately 50% of the world's proven recoverable oil and gas resources (Lin et al., 2021; Xu et al., 2020). Currently, carbonate reservoirs hold about 60% of the world's proven and probable oil reserves (Odintsova et al., 2021; Xiao et al., 2021).

However, carbonate reservoirs, especially tight carbonate gas reservoirs, generally achieved relatively low primary resource recovery rates. Most reservoirs are typically stimulated by acidizing or acid fracturing treatment. The acid reacts with the carbonate formation to create a conductive path for the hydrocarbon to flow. The main reason for the low efficiency of most acid treatments of heterogeneous reservoirs is the irregular penetration of injected fluids into the formation, primarily flowing into the most permeable zones. In turn, the less permeable zones are barely exposed to the injected acid (Gabzalilova et al., 2021). To prevent this, highly permeable zones need to be isolated temporarily for the duration of acid injection (Folomeev et al., 2021). Effective diversion is the key for the success of carbonate stimulation treatments, especially for long horizontal and multi-lateral wells (Gomaa et.al 2010). There are mainly two types of diversion techniques, chemical and mechanical diversion techniques, have been used to produce a uniform distribution of acid across the contrasting permeability layers. The mechanical diverting technique uses zonal isolation tools such as packers, ball sealers or particles. However, they are either more expensive or operationally time-consuming than chemical diversion techniques and furthermore not applicable or effective in open-hole completions. Chemical diverting techniques such as foam and viscous gels are commonly used in acidizing or acid fracturing treatment of carbonate reservoirs. The foams are operationally complicated and reduce the acid strength of the treatment fluids even though they are effective in diversion. Viscoelastic surfactant-based gelled acids are also used with effective diversion and less damaging, but they are generally much more costly than gelled acids, which is more often used in acidizing or acid fracturing treatment of carbonate reservoirs.

Non-iron crosslinked self-diverting gelled acids are in-situ crosslinked polymeric gelled acids, which consists of HCl, a polymer, a crosslinker, and breaker. Partially hydrolyzed polyacrylamides have carboxylic groups that crosslink with iron or zirconium at pH 2-4 (Deysarkar et al. 1984). Initial spending of the live acid raises the pH to above 2, at which polymer is crosslinked and viscosity of the spent acid increases rapidly. The highly viscous acid will reduce the acid flow rate into the treated zones, forcing the live acid to flow into the untreated zones. At pH 4, the breaker is active and breaks the viscous acid into a low viscosity fluid (Magee et al. 1997; Hill 2005).

One of the concerns about iron crosslinked self-diverting gelled acids is the presence of ferric ions in the system. It is well known that iron ions can precipitate in the formation and cause formation damage. For example, Taylor et al. (1999) showed that the type of iron precipitate depends on the level of hydrogen sulfide present. In sweet wells (no hydrogen sulfide), iron hydroxides will precipitate at pH range from 1 to 2. Crowe (1985) and Brezinski (1999) showed that iron sulfide species are formed at pH 1.9 in sour gas wells. Acid sludge or emulsions and more corrosion problems are also reported in the presence of ferric ions in carbonate stimulation treatment using iron crosslinked self-diverting gelled acids.

In addition, the chains of anionic polyacrylamide contain negatively charged functional groups such as carboxyl ($-COO^-$). In strong acid such as hydrochloric acid, proton H^+ reacts with the carboxyl group to form a neutral carboxylic acid ($-COOH$), which reduces the charge density of the anionic group. This neutralization effect weakens the electrostatic repulsion between the polymer chains, causes curling

Downloaded from https://onepetro.org/SPEMEOS/25MEOS/conference/25MEOS by JSC Zarubezhneft, Alexey Solovyev on 07 November 2025

and aggregation of the originally extended polymer chains, and thereby reduces the polymer solubility. Meanwhile the reaction product $CaCl_2$ has adverse effect on the viscosity of partially spent acids and therefore, Gomaa and Nasr-El-Din (2010a) recommended using 3-5 wt% HCl based self-diverting gelled acid for diversion. However, low concentration hydrochloric acid severely limits the efficiency of acidizing treatment.

In order to avoid the anionic polymer hydration problem, minimize the secondary formation damage introduced by iron ions, and eliminate the impact of calcium chloride on the viscosity of spent acid in traditional iron crosslinked self-diverting gelled acids, we introduced a novel chemical mechanism including both base polymer and crosslinker to prepare the novel non-iron crosslinked self-diverting gelled acids. The novel non-iron crosslinked self-diverting gelled acids are comprehensively evaluated in acidizing laboratory in terms of fluid stability, iron control capacity, corrosion inhibition performance, and rheological behavior at temperatures up to 275°F.

Experimental

Materials and Equipment

36.9% HCl (ACS reagent grade) was used as base acid for fluid preparation. Calcium carbonate powder (ACS grade) was used to neutralize live acids. Polymer and other additives used in the fluid system are listed in Table 1.

Table 1—Experimental Materials and Fluid Formulation

Description	Form	Fluid Formulation, gpt or ppt
Hydrochloric Acid, ACS grade	Liquid	503.8
Calcium Carbonate, ACS grade	Solid	/
Flowback surfactant	Liquid	2
Iron reducing agent	Solid	50
NISGA crosslinker	Solid	53
pH Buffer	Liquid	2
Acid gelling agent slurry	Liquid	18.2
HT inhibitor aid	Solid	7
Corrosion inhibitor	Liquid	20
Encapsulated breaker	Solid	0-8

In addition to regular laboratory apparatus such as glass wares, thermometer, pH meter and water bath, special laboratory equipment are utilized in this project to evaluate the performance of the non-iron crosslinked self-diverting gelled acids. The equipment include high-temperature high-pressure rheometer and corrosion cell. Specification of the laboratory equipment are described in Table 2.

Table 2—Experimental Equipment and Apparatus

Name	Function	Model
Viscometer	viscosity measurement	Fann35A
HTHP Rheometer	fluid rheology characterization	C5550
HTHP Corrosion Cell	corrosion characterization	SY-FSY
ICP-AES	element analysis	HK-8100

Downloaded from https://onepetro.org/SPEMEOS/25MEOS/conference/25MEOS by JSC Zarubezhneft, Alexey Solovyev on 07 November 2025

Experimental Procedures

Fresh Acid Preparation. Measure the required amount of water and transfer it to a beaker. Stir with overhead blender and adjust the speed to about 1000 rpm. Add the required amount of 36.9% HCl to the above solution and continue mixing for about 2 minutes. Add the polyacrylamide slowly and continue to blend for 30 minutes. Add corrosion inhibitor aid and iron control agent to the solution and continue to mix for 5 minutes. Finally add corrosion inhibitor, flowback surfactant and crosslinker to the solution and continue to blend for 5 minutes.

Spent Acid Preparation. Measure and add 150 cm^3 of fresh acid into a 250 cm^3 glass bottle with lid. Place the glass bottle with the lid in water bath that has been preheated to 194°F. Timer 10 minutes from the time it is placed. Transfer the acid from the glass bottle to 1000 cm^3 beaker and place it in water bath that has been preheated to 158°F. Small amount of $CaCO_3$ powder is added by multiple times while stirring with a glass rod, and the pH is monitored using a pH meter. When the pH reaches 1-2, a significant increase in viscosity is observed.

Stability of Fresh Gelled Acids. For room temperature stability test, viscosity of fresh acid was first measured at 511 s^{-1} with Fann35A at 95°F. The acid solution is then placed in a glass bottle with lid and allowed to age at 95°F. After 24 hours and 48 hours aging, the viscosity was measured again with Fann35A at 95°F.

Regarding high-temperature stability test, viscosity of fresh acid was measured at 511 s^{-1} with Fann35A at 77°F, Transfer the fresh acid into a high-temperature glass bottle and then place them into a high-temperature high-pressure heating assembly with cap sealed properly. Pressurize the heating assembly at 500 psi. Heat the whole assembly in the oven at test temperature for 3 hours. Take out the assembly from the oven and cool down to room temperature. Dissemble the assembly and measure the acid viscosity again with Fann35A at 77°F.

Rheological Properties of Spent Gelled Acids. The viscosity of the spent gelled acid at various pH is tested using the Fann35A at ambient conditions. The spent acids are prepared as per procedures described in the section "Spent Acid Preparation".

Rheological properties of gelled acids at high-temperature and high-pressure are measured using Chandler 5550. The fluid is heating up from room temperature to target temperature in 20 minutes. The total testing duration including 20 minutes heating is about 60 minutes and the shear rate is 100 s^{-1} across the whole test duration.

Iron Control. Prepare 294 cm^3 19 wt.% $CaCl_2$ solution and mix 6 cm^3 HCl into the $CaCl_2$ solution. Weigh 4.356 grams $FeCl_3.6H_2O$ and dissolve into the above solution. The iron concentration of the initial solution is measured by ICP-AES. Add iron control agent into the solution while mixing and then slowly add 5 wt. % Na_2CO_3 solution until the solution pH reaches 4. Cure the fluid at 275°F for 3 hours and measure the iron concentration of the fluid before and after aging using ICP-AES.

Corrosion Control. The corrosion tests were carried out using C95 and Q125 coupons in 15% HCl at 225°F, 250°F and 275°F. Clean and dry coupons are used. Pour the prepared acid into the kettle at acid volume to coupon area ratio of 20 cm^3/cm^2 and then hang up the coupon assembly in the corrosion cell. Pressurize the corrosion cell. Set temperature, heating rate and test duration 4 hours and start testing. Turn off the equipment upon completion of the test and cool down to below 104°F. Drain the acid and take out the coupons. Calculate the corrosion rate and record the pitting index after washing, drying and weighing the coupons.

$$Corrosion\ Rate\ \left(lbs/ft^2\right) = \frac{Weight\ loss\ in\ grams\ \times 2.048}{Coupon\ area\ in\ cm^2}$$

Downloaded from https://onepetro.org/SPEMEOS/25MEOS/conference/25MEOS by JSC Zarubezhneft, Alexey Solovyev on 07 November 2025

Results and Discussions

Stability of Fresh Gelled Acids

The fresh acid is prepared as per procedures described above and formulation listed in Table 1. Viscosity of the fresh acid is measured at 511 s^{-1} with Fann35A at 95°F to evaluate the gelled acid stability at room temperature. The results are listed below in Table 3.

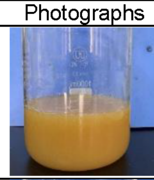
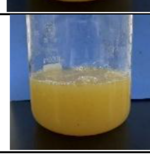
Table 3—Stability of Fresh Acids at Room Temperature

Fresh Acid	Fresh Acid Initially Prepared	Fresh Acid Aged for 24 hours at 95°F	Fresh Acid Aged for 48 hours at 95°F
Viscosity at 511 s^{-1}	30 cP	29 cP	28 cP

The stability results indicate that fresh gelled acids are practically (95°F for at least 48 hours) stable at regular field application conditions.

Viscosity of fresh gelled acid is measured at 511 s^{-1} with Fann35A at 77°F before and after aging at high-temperature and high-pressure to evaluate the fresh gelled acid stability at elevated field application conditions. The results are listed below in Table 4.

Table 4—Stability of Fresh Acids at Elevated Temperature and Pressure

Aging	Before Aging		After Aging	
	Photographs	Viscosity	Photographs	Viscosity
275°F for 3 hours		40 cP		23 cP
250°F for 3 hours		32 cP		17 cP
225°F for 3 hours		30 cP		14 cP

The high-temperature and high-pressure stability results show that the viscosity of the fresh gelled acids at 77°F and 511 s^{-1} is reduced partially after aging. However, the gelled acids were still clear solution and no precipitations observed, indicating that no adverse change or reaction takes place while pumping the fresh gelled acids through the wellbore tubulars and accessories.

Rheological Properties of Spent Gelled Acids

Viscosity of Gelled Acids at Various pH and Self-diverting Mechanism. Spent gelled acids are prepared using different amounts of $CaCO_3$ to neutralize the fresh gelled acids into spent gelled acids at various pH as per procedures described above. After cooling the gelled acids down to room temperature, the viscosity of the spent gelled acids is measured using Fann35A viscometer. The results are illustrated in Fig. 1.

Downloaded from https://onepetro.org/SPEMEOS/25MEOS/conference/25MEOS by JSC Zarubezhneft, Alexey Solovyev on 07 November 2025

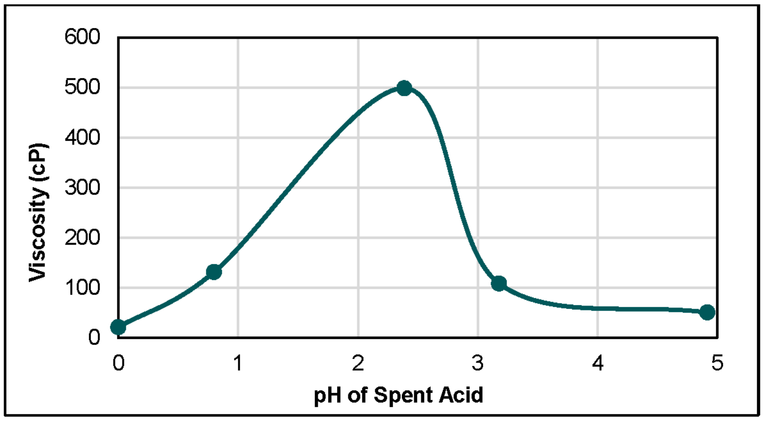


Figure 1—Viscosity of Spent Acids at Various pH

Polymer chains contain amide groups. H^+ in acids can protonate the carbonyl oxygen on the amide group as temperature rises, forming a positively charged intermediate (-$CONH_2^+$). Water molecules act as nucleophiles to attack the protonated carbonyl carbon, resulting in a tetrahedral intermediate. The tetrahedral intermediate breaks the amide bond and generates a carboxylic group. On the other hand, cationic ions presented in novel non-iron self-diverting gelled acids can form a complex structure that acts as a crosslinker to react with carboxylate or hydrolyzed groups on the polymeric chains to form a crosslinking network. The crosslink reaction is sensitive to pH and occurs within a narrow pH range between 1 and 3. Moreover, it is not possible chemically for special cationic crosslinker to form precipitation or sludge across carbonate acidizing process (pH range from negative to 7). This unique chemical mechanism eliminates the secondary precipitations and formation damage introduced by iron crosslinked self-diverting gelled acids.

The viscosity profile illustrated in Fig. 1 indicates the self-diverting performance of the non-iron crosslinked self-diverting gelled acids. Fresh gelled acid viscosity is low in the beginning to facilitate the field mixing and job execution. Viscosity of the spent gelled acid increases while fresh gelled acids contact and react with high permeability carbonate zones. The high viscosity of spent gelled acids will create flow resistance in the high permeability zone and divert the fluid flow into the low permeability zone. The higher is the spent gelled acid viscosity, the better is the self-diverting performance of the gelled acids. The acids will gradually flow back to high permeability zones while spent acid pH decreases as illustrated in Fig. 1 because more and more acids react with carbonate formation, leading to the self-diverting behavior of the novel non-iron crosslinked gelled acids.

Rheological Properties of Spent Gelled Acids at High-temperature and High-pressure. Take the spent acid prepared above at pH 2 and conduct high-temperature high-pressure rheological testing using Chandler 5550. The results are illustrated below in Fig. 2. Viscosity of spent acids at pH 2 is more than 400 cP at 100 s^{-1} at 275°F and 500 psi after 60 minutes (yellow curve in Fig. 2), indicating the high diversion performance of the polymeric self-diverting gelled acids at reservoir conditions.

Downloaded from https://onepetro.org/SPEMEOS/25MEOS/conference/25MEOS by JSC Zarubezhneft, Alexey Solovyev on 07 November 2025

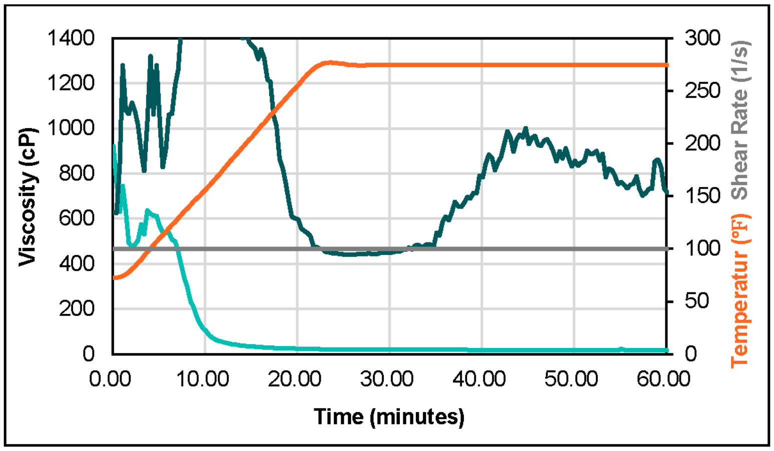


Figure 2—Rheological Property of Spent Acids with and without Breaker at 275°F and 500 psi

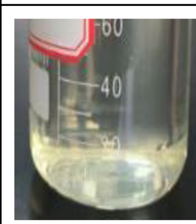
As we know, there is risk of formation damage if the gelled acids are not broken and flow back after carbonate acidizing or acid fracturing. Therefore, we introduced an internal breaking mechanism in the novel non-iron crosslinked self-diverting gelled acids in addition to the low viscosity profile of spent acids after acidizing or acid fracturing. The viscosity of the spent acids drops to less than 20 cP at 100 s^{-1} at 275°F and 500 psi after 60 minutes (blue curve in Fig. 2), indicating the spent gelled acids are broken completely to flow back and non-damaging to the post-treatment formation or fractures.

Iron Control

Selection of iron control agents is challenging for crosslinked self-diverting gelled acids. Chelating agents are commonly used for iron control in iron-crosslinked self-diverting gelled acids. However, they are not applicable for novel non-iron crosslinked self-diverting gelled acids because cationic crosslinker in the gelled acids is not compatible with chelating agents. A novel iron control agent is developed in this project to prevent iron hydroxides precipitation.

Up to 3000 ppm ferric ions are added into the fresh gelled acid and iron concentration is analyzed by ICP-AES. Physical appearance (brownish clear solution) and iron concentration (3015 ppm) in fresh gelled acid are listed in the left column in Table 5. Clear gelled acid (light brownish solution) and iron concentration (2987 ppm) were obtained right after neutralization of the fresh gelled acid as shown in the middle column in Table 5. Photographs (clear light-yellow solution) of the spent gelled acid after aging at 275°F for 3 hours was also taken and iron concentration was also measured (2895 ppm) as shown in the right column in Table 5. The results described above conclude that the iron control agent selected for the novel non-iron crosslinked self-diverting gelled acids can control more than 3000 ppm ferric ions in both fresh and spent gelled acids at 275°F.

Table 5—Iron Control Test Results

	Before Neutralization	After Neutralization	After Aging at 275°F for 3 Hours
photographs			
Iron Concentration	3015 ppm	2987 ppm	2895 ppm

Downloaded from https://onepetro.org/SPEMEOS/25MEOS/conference/25MEOS by JSC Zarubezhneft, Alexey Solovyev on 07 November 2025

Corrosion Control

Normally it is more difficult to control corrosion of crosslinked self-diverting gelled acids than regular acids. Therefore, the selection of corrosion inhibitor and inhibitor aid is critical for development of non-iron crosslinked self-diverting gelled acids.

Static corrosion (typically the worst scenario) tests were conducted at temperature range 225°F to 275°F and 2000 psi. Total test duration is 6 hours including 1 hour heating and 1 hour cooling process. Corrosion control performance (corrosion rate ranges between 0.007 and 0.028 lbs/ft^2 and all pitting index results are less than 1) of the novel non-iron crosslinked self-diverting gelled acids are illustrated in Tables 6 and 7. Results indicate that the combination package of corrosion inhibitor and inhibitor aid selected in this project is compatible with other additives in the novel non-iron crosslinked self-diverting gelled acid, and corrosion rate and pitting index meet the mostly-used industry criteria (corrosion rate < 0.05 lbs/ft^2 and pitting index < 1) for both C95 and Q125 coupons in gelled acids up to 275°F for 6 hours.

Table 6—Corrosion Test Results with C95 Coupons

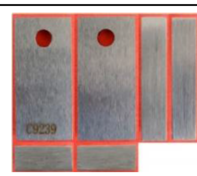
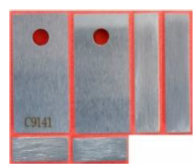
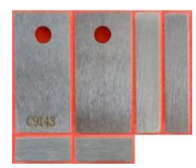
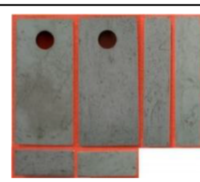
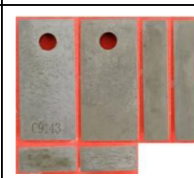
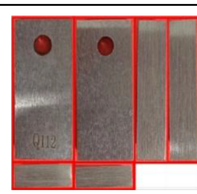
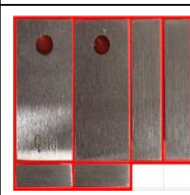
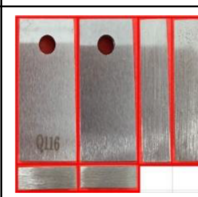
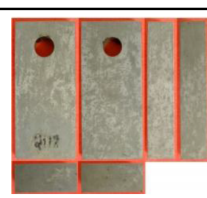
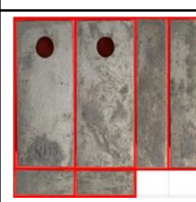
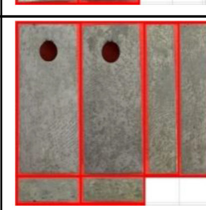
Test Temperature	275°F	250°F	225°F
Test Condition	0 rpm, 2000 psi, 4 hrs.	0 rpm, 2000 psi, 4 hrs.	0 rpm, 2000 psi, 4 hrs.
Duration, hrs. (heating/constant temp./cooling)	6:00 (1/4/1)	6:00 (1/4/1)	6:00 (1/4/1)
Corrosion Rate, lbs/ft^2	0.026	0.013	0.008
Pitting Index	<1	<1	<1
Coupon Photographs before Corrosion Test			
Coupon Photographs after Corrosion Test			

Table 7—Corrosion Tests Using Q125 Coupon

Test Temperature	275°F	250°F	225°F
Test Condition	0 rpm, 2000 psi, 4 hrs.	0 rpm, 2000 psi, 4 hrs.	0 rpm, 2000 psi, 4 hrs.
Duration, hrs. (heating/constant temp./cooling)	6:00 (1/4/1)	6:00 (1/4/1)	6:00 (1/4/1)
Corrosion Rate, lbs/ft^2	0.028	0.015	0.007
Pitting Index	<1	<1	<1
Coupon Photographs before Corrosion Test			
Coupon Photographs after Corrosion Test			

Downloaded from https://onepetro.org/SPEMEOS/25MEOS/conference/25MEOS by JSC Zarubezhneft, Alexey Solovyev on 07 November 2025

Conclusions

1. A novel non-iron crosslinked self-diverting gelled acid is successfully developed. The novel system include the base polymer, crosslinker, iron control agent and corrosion inhibitor and inhibitor aid.
2. Laboratory test results indicate the good performance of the novel non-iron crosslinked self-diverting gelled acid in terms of fresh acid stability, rheological properties of the spent acids, iron and corrosion control capacities.
3. The fresh gelled acid is stable at least 48 hours at 95°F and after aging at 275°F for 3 hours, indicating that the novel gelled acids are applicable in most oilfield operation conditions.
4. The rheological profile at various pH and shear stability at high temperature and pressure illustrate the superior self-diverting performance of the novel non-iron crosslinked self-diverting gelled acid. An internal breaker was also developed in this project to ensure the complete breaking of spent gelled acid and minimize the potential formation or fracture damage.
5. An iron control agent is specifically developed for the novel self-diverting gelled acid system. It can control up to 3000 ppm ferric ions in both fresh and spent gelled acids.
6. The combination of corrosion inhibitor and inhibitor aid selected for the novel acid systems effectively controls the corrosion rate and pitting index of the acids up to 275°F for C95 or Q125 tubular materials.

Acknowledgements

The authors want to acknowledge management of TAQA and OPT companies for the permission to publish subject manuscript. We also thank Saudi Aramco Management for supporting this project.

References

Lin L.; Liu S.; Xua Y.; Lia F. 2021. Lithofacies and Pore Structure of Fine-grained Sedimentary Rocks of Qing-1 Member of Cretaceous in the Southern Songliao Basin. Paper presented in *ACTA GEOLOGICA SINICA*, 2021, **95**, 2, 570–584. https://doi.org/10.1111/1755-6724.14657

Xu Z.; Li, S.; Li, B.; Chen D.; Liu Z.; Li, Z. 2020. A review of development methods and EOR technologies for carbonate reservoirs. Paper presented in *Petroleum Science*, 2020, **17**, 990–1013. https://doi.org/10.1007/s12182-020-00467-5

Odintsova, A.A.; Rybkina, A.I.; Nikolova, J.I.; Korolkova, A.A. 2021. ROSA database and GIS project: accumulation of the world largest oil and gas deposits in geological history. 337–350. In: Svalova, V. (Ed), *Heat-Mass Transfer and Geodynamics of the Lithosphere. Innovation and Discovery in Russian Science and Engineering*. Springer, Cham. https://doi.org/10.1007/s12182-020-00467-5

Xiao D.; Cao J.; Tan X.; Xiong Y.; Zhang D.; Dong G.; Lu Z. 2021. Marine carbonate reservoirs formed in evaporite sequences in sedimentary basins: A review and new model of epeiric basin-scale moldic reservoirs. Paper presented in *Earth-Science Reviews*. 2021, **223**, 103860. https://doi.org/10.1016/J.EARSCIREV.2021.103860

Gabzalilova A.K.; Batalov D.A.; Mukhametshin V.S.; Andreev V.E. 2021. Geological and technological justification of the parameters of acid-clay treatment of wells. Paper presented in IOP Conf. Ser.: Mater. Sci. Eng. 2021, **1064**, 012058. https://doi.org/10.1088/1757-899X/1064/1/012058

Folomeev A.; Taipov I.; Khatmullin, A., et al 2021. Acid vs. self-diverting systems for carbonate matrix stimulation: An experimental and field study. Paper presented at the SPE Russian Petroleum Technology Conference, Virtual, October 2021. Paper Number: SPE-206647-MS. https://doi.org/10.2118/206647-MS

Gomaa, A.M. 2010. Proper design of polymer-based in-situ gelled acids to improve carbonate reservoir productivity. Paper presented at the SPE Annual Technical Conference and Exhibition, Florence, Italy, September 2010. Paper Number: SPE-141123-STU. https://doi.org/10.2118/141123-STU

Deysarkar, A.K.; Dawson, J.C.; Sedillo, L.P.; Knoll-Davis, S. 1984. Crosslinked Acid Gel. Paper presented in *J Can Pet Technol.* 1984, **23**, 01, 24–26. https://doi.org/10.2118/84-01-01

MaGee J.; Buijse M.A.; Pongratz R. 1997. Method for Effective Fluid Diversion when Performing a Matrix Acid Stimulation in Carbonate Formations. Paper presented at the Middle East Oil Show and Conference, Bahrain, March 1997. Paper Number: SPE-37736-MS. https://doi.org/10.2118/37736-MS

Hill D.G. 2005. Gelled Acid. United States Patent Application Publication. US patent US2005/0065041 A1, issued on March 24, 2005.

Downloaded from https://onepetro.org/SPEMEOS/25MEOS/conference/25MEOS by JSC Zarubezhneft, Alexey Solovyev on 07 November 2025

Taylor K.C.; Nasr-El-Din H.A.; Al-Alawi M.J. 1999. Systematic Study of Iron Control Chemicals Used During Well Stimulation. *SPE J.* 1999, **4** (01): 19–24. Pater Number: SPE-54602-PA. https://doi.org/10.2118/54602-PA

Crowe C.W. 1985. Evaluation of Agents for Preventing Precipitation of Ferric Hydroxide From Spent Treating Acid. *J Pet Technol.* 1985, **37** (04): 691–695. Paper Number: SPE-12497-PA. https://doi.org/10.2118/12497-PA

Brezinski M.M. 1999. Chelating Agents in Sour Well Acidizing: Methodology or Mythology. Paper presented at the SPE European Formation Damage Conference, May 31–June 1, 1999. Paper Number: SPE-54721-MS. https://doi.org/10.2118/54721-MS

Nasr-El-Din H.A.; Taylor K.C.; Al-Hajji H.H. 2002a. Propagation of Cross-Linkers Used in In-Situ Gelled Acids in Carbonate Reservoirs. Paper presented at the SPE/DOE Improved Oil Recovery Symposium, April 13-17, 2002. Paper Number: SPE-75257-MS. https://doi.org/10.2118/75257-MS

Nurlybayev, N., Z. Al-Jalal, M. Samuel, et al 2023. Addressing Hydraulic Fracturing Performance Challenges in HPHT Well with New Fluid System and Comprehensive Fracturing Study Utilizing Full 3D Simulator. Paper presented at the *Middle East Oil, Gas and Geosciences Show*, Manama, Bahrain, February 2023. Paper Number: SPE-213494-MS. https://doi.org/10.2118/213494-MS

Samuel, M., Al-Jalal, Z., Nurlybayev, N., et al 2023. Novel Carbon Dioxide (CO2) Foamed Fracturing Fluid, an Innovative Technology to Minimize Carbon Footprint. Paper presented at the *Middle East Oil, Gas and Geosciences Show*, Manama, Bahrain, February 2023. Paper Number: SPE-213453-MS. https://doi.org/10.2118/213453-MS

Yudin, A., M. Farouk; Z. Al-Jalal, et al 2024. Optimization of Fluid Lab Testing for Conventional Proppant Fracturing Applications in HT Reservoirs. Paper presented at the International Petroleum Technology Conference, Dhahran, Saudi Arabia, February 2024. Paper Number: IPTC-24398-MS. https://doi.org/10.2523/IPTC-24398-MS

Yudin, A., Nurlybayev, N., Al-Jalal, Z. 2023. Solving the Low Injectivity Challenges in Hydraulic Fracturing Tight Gas Reservoirs - Case Histories Review. Paper presented at the *Middle East Oil, Gas and Geosciences Show*, Manama, Bahrain, February 2023. Paper Number: SPE-213424-MS. https://doi.org/10.2118/213424-MS

Nurlybayev, N., Yudin, A., et al 2024. A Comprehensive Study of the Various Stimulation Fluids Deployed in HPHT Reservoirs. Paper presented at SPE Caspian Technical Conference and Exhibition scheduled to be held in Atyrau, Kazakhstan, 26 – 28 November 2024. Paper Number: SPE-223439-MS. https://doi.org/10.2118/223439-MS

Downloaded from https://onepetro.org/SPEMEOS/25MEOS/conference/25MEOS by JSC Zarubezhneft, Alexey Solovyev on 07 November 2025